

Matthew Arboleda

Los Angeles, California

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Website: <https://ma0610.github.io/LatestPortfolioWebsite/>

SKILLS

- **Languages/Frameworks:** Java, Python, PyTorch, Flask, HTML, CSS, Javascript, React, MySQL, SQLAlchemy, Linux, Windows, Git
- **Tools:** Render, Cloudflare, Figma

RELEVANT COURSEWORK

- Data Structures, Computer Organization, Math Foundations of Computer Science, Human-Computer Interaction, Computer Architecture, Practicum in Computer Science, Deep Learning, Social Data Science

EDUCATION

Occidental College

Expected Graduation: May 2026

Cumulative GPA: 3.76/4.0

Bachelor of Arts in Computer Science Major

CERTIFICATES

- **IBM Developing Front-End Apps with React 2025**

EXPERIENCE

Occidental Research Project: ELPAC Phoneme Detection App Jan 2026 - Present

- Developed Android application leveraging Wav2Vec 2.0 phoneme model to provide real-time speech analysis for ELPAC test preparation
- Integrated pre-trained phoneme recognition model with Android microphone input to enable automatic phoneme detection and feedback
- Designed tool to assist teachers in identifying student pronunciation gaps and tracking progress on targeted phonemes

AlgoVerse AI Research Jul 2025 - Oct 2025

- Conducted research on multi-agent system security vulnerabilities using encryption-based jailbreak techniques across OpenAI, Anthropic, and Gemini LLMs

Teacher's Assistant, Los Angeles, CA Feb 2025 - Present

- Graded and provided detailed written feedback for 80+ problem sets in Math Foundations of Computer Science course at Occidental College

Scheduling Website, Los Angeles, CA Aug 2024 - Dec 2024

- Led team to design a course scheduling website for Occidental Biochemistry department, enhancing inter-department communication
- Developed front end logic and SQLAlchemy database schema to facilitate data transfer
- Implemented Python Flask to ensure functionality according to departmental needs

AWARDS & SCHOLARSHIPS

Occidental Honor Roll Aug 2022 - Present

- Achieved a Cumulative GPA of 3.5 or higher

Cosmos Scholar (NSF-funded scholarship program for STEM students) Jun 2023 - Present

- Exemplified academic talent in STEM courses

References Available on Request